



# Emotional Load as a Boundary Condition in the Workload–Workplace Aggression Relationship: Evidence From an Industrial Firm

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**Abstract:** Workplace aggression poses persistent challenges to employee well-being and organizational functioning, particularly in industrial settings where employees face simultaneous task and emotional demands. However, limited evidence is available on whether emotional load changes the strength of the association between workload and different forms of workplace aggression. This study investigates the moderating role of emotional load in the relationships between workload and psychological and physical aggression. A cross-sectional survey was conducted among 275 employees of an industrial firm in Iran. Workload, emotional load, and workplace aggression were measured using established scales. Self-reported and coworker-reported measures of workload and aggression were included. Pearson correlation analysis and hierarchical multiple regression were performed, with mean-centered interaction terms entered to test the proposed moderating relationships. The results showed that workload was positively correlated with psychological and physical aggression. Significant workload × emotional load interactions were found in the models predicting self-reported psychological and physical aggression ( $p < 0.01$ ), whether workload was evaluated by employees or coworkers. These interaction terms explained an additional 1%–3% of the variance. However, the corresponding interactions were not consistently significant when aggression was rated by coworkers. The findings suggest that emotional load serves as a boundary condition in the relationship between workload and self-reported aggressive behavior, although this pattern is less consistent in coworker reports. The study clarifies how concurrent task and emotional demands are jointly associated with employee aggression and provides industrial managers with a basis for addressing emotional demands alongside workload when developing workplace aggression prevention measures.

**Keywords:** Workload; Emotional load; Workplace aggression; Employee behavior; Job demands; Moderation analysis; Industrial workplace

## 1. Introduction

Workplace aggression is a persistent concern in organizational research because it threatens employee well-being as well as organizational performance. A recent large-scale meta-analysis covering 471 samples from 36 countries showed that experienced workplace aggression reduced task performance and organizational citizenship behavior while increasing deviant behavior, primarily through psychological strain and negative affect (Zhong et al., 2025). Despite extensive evidence concerning its consequences, less is known about why employees exposed to comparable levels of work pressure differ in their aggressive responses. This study addresses this question by examining whether the emotional demands experienced at work shape the relationship between workload and workplace aggression.

This question has become particularly relevant in contemporary industrial workplaces. Digitalization, automation, and technology-driven work processes have changed the pace, coordination, and organization of industrial work. Automated systems, digital monitoring tools, and real-time performance tracking have altered

both the nature of employee workload and the conditions under which work is performed (Parker & Grote, 2022; Siegel et al., 2022). Employees in technology-intensive industrial settings may face constant connectivity, accelerated workflows, and greater performance visibility, all of which can increase cognitive demands and emotional pressure (Chowdhury et al., 2024; Frey & Osborne, 2017). Electronic performance monitoring may improve operational efficiency, but it can also create perceptions of surveillance and contribute to stress and emotional exhaustion (Nawaz et al., 2024; Stanton & Weiss, 2000). Although the present study does not treat digitalization or technology use as separate explanatory variables, these changes provide an important organizational context for understanding how workload and emotional demands jointly relate to employee behavior.

Workplace aggression includes verbal hostility, rumor spreading, harassment, sabotage, theft, and, in more serious cases, physical confrontation. A foundational typology defines these behaviors as voluntary acts that violate organizational norms and threaten the organization or its members (Robinson & Bennett, 1995). Although workplace aggression can arise from different individual and organizational conditions, it rarely occurs without a precipitating factor. A substantial body of research identifies organizational stressors, particularly excessive workload, as important antecedents of aggressive and counterproductive behavior (Fida et al., 2014; Meier & Spector, 2013). Workload alone, however, does not explain why employees facing similar demands respond differently.

Emotional load may account for part of this variation. It refers to the emotional demands employees encounter while performing their work, including emotionally difficult interactions, feelings of personal threat, and the effort required to manage work-related emotions. Existing research has generally examined workload and emotional demands as separate predictors of employee outcomes, leaving their joint relationship with workplace aggression less clearly understood. A recent study of the interplay between mental load and emotional load found that the relationship between mental load and counterproductive behavior changed according to the level of emotional load: mental load was associated with greater engagement when emotional load was low but with greater deviance when emotional load was high (Machek & Machek, 2025). This evidence supports treating emotional load not merely as an additional job demand, but as a boundary condition that may determine when workload is more strongly associated with aggressive behavior (Machek & Machek, 2025).

The consequences of this relationship are both behavioral and economic. Emotional labor, when combined with exposure to workplace violence, has been found to predict health-related productivity loss through both absenteeism and presenteeism in a large nationwide workforce sample (Ko et al., 2024). The combined influence of workload and emotional load is therefore relevant not only to employee conduct and well-being but also to organizational functioning. Identifying the conditions under which workload is more strongly associated with aggression may help organizations design more focused measures for managing job demands and preventing harmful workplace behavior.

Against this background, the present study investigates whether emotional load moderates the relationships between workload and psychological and physical aggression among employees of an industrial firm in Iran. The study distinguishes between psychological and physical aggression because these behaviors differ in their form and frequency and may not respond to work-related demands in the same way. It also considers both self-reported and coworker-reported assessments, allowing the proposed relationships to be examined across different reporting sources. Accordingly, the following hypotheses are proposed:

Hypothesis 1 (H1): Emotional load moderates the relationship between workload and psychological aggression, such that the positive association between workload and psychological aggression is stronger when emotional load is high.

Hypothesis 2 (H2): Emotional load moderates the relationship between workload and physical aggression, such that the positive association between workload and physical aggression is stronger when emotional load is high.

This study makes three contributions. First, it treats emotional load as a boundary condition in the workload–aggression relationship rather than solely as an independent job demand. Second, it examines whether this moderating relationship applies to both psychological and physical aggression. Third, by considering self-reported and coworker-reported assessments, it examines whether the observed relationships remain consistent across reporting sources. These contributions extend research on job demands and employee behavior while providing evidence relevant to workload management and aggression prevention in industrial workplaces.

The remainder of the paper is organized as follows. Section 2 reviews the theoretical mechanisms linking workplace stressors to aggression and positions emotional load within this relationship. Section 3 describes the sample, measurement instruments, and analytical approach. Section 4 reports the correlation and moderation results. Section 5 discusses the findings in relation to previous research and considers their practical implications, and Section 6 presents the conclusions and limitations of the study.

## 2. Background

### 2.1. Workplace Aggression and Counterproductive Work Behavior

Research interest in unethical and counterproductive employee behavior has grown substantially, partly because such behavior is associated with increased insurance claims, higher injury rates, and reduced productivity (Christian & Ellis, 2014; LeBlanc & Kelloway, 2002). These organizational costs have also been documented across different forms of productive and counterproductive work behavior (Coyne et al., 2013; LeBlanc & Kelloway, 2002; Vigoda, 2002). Workplace aggression refers to behavior intended to harm an organization or one or more of its members (Fox & Spector, 1999). It falls within the broader domain of counterproductive work behavior, which includes theft, sabotage, verbal abuse, deception, and the deliberate withdrawal of effort (Spector & Fox, 2002). One of the most widely adopted frameworks classifies these behaviors according to whether they are directed at individuals or the organization and whether their severity is minor or serious (Bennett & Robinson, 2000; Robinson & Bennett, 1995).

Within this framework, psychological aggression includes actions such as verbal hostility, humiliation, intimidation, and social undermining, whereas physical aggression involves direct or threatened physical harm. Although both forms violate organizational norms, they differ in their expression, frequency, and potential consequences. Examining them separately therefore provides a more precise account of how workplace demands are associated with aggressive employee behavior.

### 2.2. Theoretical Foundations of the Workload–Aggression Relationship

Two theoretical perspectives explain how organizational conditions may give rise to workplace aggression. The stressor-emotion model, which draws on the frustration–aggression framework, proposes that job stressors generate negative emotional states that may lead to counterproductive behavior as a coping response or an expression of accumulated emotion (Fox & Spector, 1999; Spector & Fox, 2002). Affective Events Theory complements this explanation by proposing that emotionally significant workplace events shape affective states, which subsequently influence employee attitudes and behavior (Weiss & Cropanzano, 1996). Taken together, these perspectives suggest that the relationship between a job stressor and an aggressive response may depend on the emotional demands simultaneously experienced by the employee. Emotional load may therefore strengthen the association between workload and aggression by reducing the emotional resources available for regulating behavioral responses.

Meta-analytic evidence supports both the general relationship between workplace stressors and harmful behavior and the role of emotional processes in this relationship. A 20-year meta-analysis of workplace incivility found that environmental stressors, individual differences, and dispositional characteristics were consistent antecedents of incivility, while negative affect represented an important pathway through which incivility produced harmful outcomes (Han et al., 2022). Evidence concerning specific workplace stressors points to a similar mechanism. Abusive supervision has been found to increase counterproductive behavior through emotional exhaustion, with the strength of this relationship reduced among employees with higher core self-evaluation (Li & Xu, 2025). These findings suggest that behavioral responses cannot be explained solely by the presence of a stressor; the employee's emotional resources and capacity to regulate negative reactions must also be considered.

#### 2.2.1 Workload and job stressors

Counterproductive and aggressive behavior has been associated with several individual and organizational factors. Relevant individual characteristics include narcissism (DeShong et al., 2015; Grijalva & Newman, 2015; Scherer et al., 2013), while organizational conditions include perceived injustice (Khan et al., 2013; Le Roy et al., 2012). A large meta-analysis of workplace aggression further identified multiple individual, interpersonal, and situational predictors of such behavior (Herscovis et al., 2007). Among job stressors, workload has received sustained attention because employees facing excessive demands may have fewer cognitive and emotional resources available to regulate their responses. Its association with counterproductive behavior has consequently been used to identify work situations in which such behavior is more likely to occur (Fida et al., 2014; Meier & Spector, 2013). Studies conducted across different occupations and research settings have consistently reported a relationship between workload or related job stressors and counterproductive behavior (Fox et al., 2001; Ilie et al., 2012; Miles et al., 2002; Penney & Spector, 2002). This relationship may also vary according to individual differences in the tendency to respond aggressively to adverse workplace conditions (Douglas & Martinko, 2001; Omar et al., 2011).

Workload can arise from several features of work design and organizational practice. Understaffing and poorly designed tasks can increase the amount and intensity of work assigned to individual employees (Cooper, 1999), while time pressure can require employees to complete tasks at a pace that places further demand on their available resources (Tan & Netessine, 2014). Excessive administrative responsibilities may add to these demands even when

they are not part of an employee's primary occupational role (Fox et al., 2001). Evidence also indicates that employees may differ in how workload is associated with burnout, including differences related to gender (Aniței et al., 2015). The relationship between workload and workplace deviance has been reported in Middle Eastern and Southeast Asian organizational samples (Bayram et al., 2009; de Jonge & Peeters, 2009; Omar et al., 2011; Radzali et al., 2013). This evidence suggests that the relationship is not confined to a single national context, although its strength and consequences may vary across cultures. Recent meta-analytic research supports this qualification by identifying individualism and masculinity as cultural moderators of the consequences of workplace aggression (Zhong et al., 2025).

The nature of workload is also changing in technology-intensive industrial environments. Digitalization and automation have introduced continuous performance monitoring, real-time productivity tracking, and faster work rhythms, potentially increasing perceived workload beyond conventional task demands (Parker & Grote, 2022; Siegel et al., 2022). Technostress refers to strain associated with the use of new technologies and may arise from technological complexity, constant connectivity, and the need to adapt repeatedly to new systems (Brod, 1984; Wang et al., 2024). Electronic monitoring and workplace surveillance may support coordination and operational control, but they can also increase perceived scrutiny, emotional exhaustion, and psychological strain (Bujold et al., 2024; Nawaz et al., 2024). Technology-related demands may consequently coexist with traditional sources of workload and create particularly demanding conditions in contemporary industrial workplaces (Malik et al., 2023; Tarafdar et al., 2007). These developments establish the organizational context of the present research; however, technology use and technostress are not included as variables in the empirical model.

### 2.2.2 Emotional load and resource depletion

Whereas workload primarily places demand on employees' time, attention, and physical or cognitive resources, emotional load concerns the emotional demands encountered in performing work. Such demands may include dealing with difficult interpersonal situations, feeling personally threatened, responding to dissatisfied individuals, and managing emotions while meeting job expectations. Exposure to workplace incivility can produce emotional and behavioral costs not only for its direct targets but also for observers and employees in lower-status positions (Porath & Pearson, 2012). Sustained emotional demands may also reduce employee well-being and increase hostile reactions when workers must continually regulate their emotions (Grandey, 2000). Emotional load can therefore operate as a hindrance demand that makes other workplace stressors more difficult to manage (Machek & Machek, 2025). When emotional load is high, employees may have fewer resources available to regulate their reactions to workload. Recent evidence accordingly indicates that manageable mental workload may be associated with engagement under low emotional load but with greater counterproductive behavior when emotional load is high (Machek & Machek, 2025). This pattern provides the theoretical basis for examining emotional load as a moderator of the association between workload and workplace aggression.

Related evidence also demonstrates the organizational relevance of emotional demands. Emotional labor combined with exposure to workplace violence has been found to predict health-related productivity loss through absenteeism and presenteeism, even after other job stressors were considered (Ko et al., 2024). Although emotional labor and emotional load are not identical constructs, both concern the emotional requirements imposed by work and the resources needed to manage them. Bullying and insufficient social support have similarly been identified as conditions associated with emotional load in the workplace (De Beer, 2014). The use of occupational well-being frameworks in South African samples further indicates that emotional and psychosocial work demands can be examined across different organizational and national contexts (Rothmann & Rothmann, 2007).

### 2.2.3 Organizational consequences of workplace aggression

The consequences of workplace aggression extend beyond the person immediately targeted. Chronic exposure to bullying is associated with deteriorating health outcomes (Hansen et al., 2006), consistent with broader findings concerning abusive behavior in the workplace (Keashly et al., 1994). Meta-analytic evidence also shows that workplace bullying produces adverse consequences at both individual and organizational levels (Nielsen & Einarsen, 2012). These consequences include damage to trust, morale, and organizational cohesion that may continue after the immediate incident has ended (Vega & Comer, 2005). At the same time, the relationship between emotional demands and counterproductive behavior can vary according to employees' self-control and perceptions of organizational justice (Bechtoldt et al., 2007; Samnani et al., 2014). Evidence from research on workplace ostracism points to a related emotional mechanism: ostracism has been found to increase counterproductive behavior through a sequential pathway involving anger and turnover intention (Zhu & Zhang, 2021). Emotional dissonance in call-center work has likewise been associated with emotional exhaustion and declining job satisfaction over time (Lewig & Dollard, 2003). Together, these findings support the broader resource-depletion logic underlying the present study: employees exposed to simultaneous task and emotional demands may have less capacity to regulate behavioral responses to workplace pressure.

### 2.3. The Present Study

Despite substantial research connecting workload, emotional demands, and workplace aggression, direct tests of emotional load as a moderator of the workload–aggression relationship remain limited. Emotional load has more often been treated as an additional predictor or component of job demands than as a condition that changes the strength of the relationship between workload and employee behavior. This limitation is particularly evident in research conducted outside North American and European settings. Recent evidence indicates that the interaction between mental load and emotional load, rather than either condition considered separately, is associated with counterproductive behavior (Machek & Machek, 2025). The present study extends this reasoning by examining psychological and physical aggression as distinct behavioral outcomes.

The industrial context also warrants attention. Employees in this sector may face a combination of physical demands, time pressure, interpersonal requirements, and adaptation to changing work processes that differs from the conditions commonly examined in service-sector and knowledge-work samples. The ongoing digitalization of industrial workplaces may add further sources of workload and emotional pressure, although these technological conditions are treated as contextual considerations rather than measured variables in this study. Evidence from an industrial workplace can therefore contribute to understanding the joint role of task and emotional demands without attributing the observed relationships directly to digitalization or technology use.

Accordingly, this study tests whether emotional load moderates the relationships between workload and psychological and physical aggression among employees of an industrial firm in Iran. It also examines these relationships using self-reported and coworker-reported assessments. This design makes it possible to determine whether the proposed moderating pattern applies across two forms of aggression and remains observable across different reporting sources. Consistent with the stressor-emotion model and Affective Events Theory, emotional load is expected to strengthen the positive association between workload and each form of workplace aggression.

## 3. Research Method

### 3.1 Research Design and Participants

A cross-sectional correlational design was adopted to examine the relationships among workload, emotional load, and workplace aggression. The study population comprised approximately 1,200 employees of an industrial firm in Iran. Using convenience sampling, 275 employees were included in the final sample. Their mean age was 36.70 years ( $SD = 11.22$ ), and their average organizational tenure was 13.35 years ( $SD = 13.11$ ). The participants worked an average of 42.82 hours per week ( $SD = 4.49$ ). The sample also included 28 employees holding managerial or supervisory positions.

The participants were informed that their responses would be used for research purposes. Participation was voluntary, and the questionnaires were completed anonymously. The study collected employee self-assessments and coworker assessments of workload and workplace aggression. Emotional load was assessed through employee self-reports.

### 3.2 Measures

#### 3.2.1. Workload

Workload was measured using the 11-item scale developed by Van Veldhoven & Meijman (1994) and subsequently used by Van Veldhoven et al. (2005). The items assessed the pace, volume, and intensity of the work performed by employees. Responses were recorded on a four-point Likert scale ranging from 1 (never) to 4 (always), with higher scores indicating greater workload. A sample item was “Do you have to work quickly to get things done?” The reliability and validity coefficients obtained for the scale were 0.87 and 0.61, respectively.

Workload was evaluated from two sources. Employees assessed their own workload, while coworkers provided corresponding evaluations of the focal employees’ workload. The two measures were analyzed separately to determine whether the proposed relationships remained consistent across reporting sources.

#### 3.2.2. Emotional load

Emotional load was measured using the seven-item scale developed by Van Veldhoven & Meijman (1994). The scale was translated for use in the present research context. It assessed emotional pressure, exposure to difficult interpersonal situations, feelings of personal threat, and the emotional demands arising from interactions at work.

The factor structure of the translated scale was examined in two subsamples drawn from the full sample. Exploratory factor analysis was first conducted using data from 102 participants. The factor loadings ranged from 0.49 to 0.82, and the extracted factor had an eigenvalue of 2.28. Confirmatory factor analysis was then performed using data from the remaining 173 participants. The analysis produced a  $\chi^2/df$  ratio of 2.24, a comparative fit index of 0.91, and a root mean square error of approximation of 0.04. The reliability and validity coefficients of the scale



were 0.84 and 0.55, respectively. These results supported the use of the translated scale in the subsequent analysis.

The seven items were as follows:

- (1) My work situations create emotional pressure for me.
- (2) I feel personally threatened in my work.
- (3) My work requires me to deal with dissatisfied or upset customers.
- (4) My work requires me to interact with and persuade other people.
- (5) Some of my colleagues occasionally make personal, non-work-related demands of me.
- (6) The emotional demands of my work are high.
- (7) Certain aspects of my work affect me personally.

The item scores were combined to produce an overall measure of emotional load, with higher scores representing greater emotional demands at work.

### 3.2.3. Psychological and physical aggression

Psychological and physical aggression were assessed using the questionnaire developed by Ménard et al. (2011). The instrument comprised two subscales: 10 items measuring psychological aggression and five items measuring physical aggression. Psychological aggression covered nonphysical forms of harmful behavior, such as verbal hostility, humiliation, intimidation, and other actions directed at undermining a coworker. Physical aggression covered behaviors involving direct or threatened physical harm.

The respondents indicated how frequently each behavior had occurred during the preceding six months. Psychological aggression was measured using a 10-point frequency scale ranging from 1 (never) to 10 (every day). Physical aggression was measured using an ordinal frequency scale ranging from 0 (not at all) to 3 (once a month or more frequently). The responses were coded so that higher scores consistently represented more frequent aggressive behavior.

Both psychological and physical aggression were evaluated through employee self-reports and coworker reports. The self-report measures represented employees' assessments of their own behavior, while the coworker-report measures represented coworkers' observations of the same forms of behavior. The two reporting sources were retained separately in the analysis.

## 3.3. Data Analysis

The data were analyzed using SPSS (Statistical Package for the Social Sciences) software. Descriptive statistics were calculated to summarize the distributions of workload, emotional load, psychological aggression, and physical aggression. Pearson's correlation coefficients were used to examine the bivariate relationships among the study variables.

Hierarchical multiple regression analysis was conducted to test whether emotional load moderated the relationships between workload and psychological and physical aggression. Before constructing the interaction terms, workload and emotional load were mean-centered to reduce nonessential multicollinearity between the main effects and their product term, following the procedures recommended by Aiken & West (1991) and Cohen et al. (2003).

Workload and emotional load were entered into the main-effects model, after which the workload  $\times$  emotional load interaction term was added. Separate regression models were estimated for psychological and physical aggression. The analyses were also conducted separately for self-reported and coworker-reported workload and aggression. A statistically significant interaction term and an increase in explained variance after its inclusion were taken as evidence of moderation. The direction of each significant interaction was interpreted by examining whether the positive association between workload and aggression became stronger at higher levels of emotional load.

## 4. Findings

### 4.1. Descriptive Statistics and Correlations

Table 1 presents the descriptive statistics for emotional load, workload, psychological aggression, and physical aggression. Workload and aggression were assessed using both self-reported and coworker-reported measures, whereas emotional load was assessed through self-reports. The descriptive results showed modest differences between the self-reported and coworker-reported measures. Coworkers reported slightly lower workload and slightly higher levels of psychological and physical aggression than the employees themselves.

The correlations among the study variables are reported in Table 2. Emotional load was negatively correlated with self-reported workload ( $r = -0.34, p < 0.01$ ) and coworker-reported workload ( $r = -0.25, p < 0.05$ ). By contrast, emotional load was positively correlated with all four measures of workplace aggression, with coefficients ranging from 0.18 to 0.34.

**Table 1.** Descriptive statistics for the study variables

| Measurement Source | Variable                 | Scale | Mean | Standard Deviation | Minimum | Maximum | Standard Error |
|--------------------|--------------------------|-------|------|--------------------|---------|---------|----------------|
| Self               | Emotional load           | 28.42 | 8.10 | 10                 | 50      | 3.23    | 0.27           |
| Self               | Workload                 | 32.52 | 0.77 | 11                 | 44      | 0.27    | 0.02           |
| Colleague          | Workload                 | 30.55 | 0.78 | 11                 | 44      | 0.27    | 0.02           |
| Self               | Psychological aggression | 1.61  | 0.53 | 1                  | 10      | 0.27    | 0.32           |
| Colleague          | Physical aggression      | 1.73  | 0.58 | 1                  | 10      | 0.29    | 0.32           |
| Self               | Physical behavior        | 1.23  | 0.34 | 0                  | 3       | 0.15    | 0.27           |
| Colleague          | Physical behavior        | 1.34  | 0.37 | 0                  | 3       | 0.16    | 0.27           |

Note: Self = self-reported measure; Colleague = coworker-reported measure.

Both workload measures were positively correlated with psychological and physical aggression. Self-reported workload was most strongly correlated with self-reported psychological aggression ( $r = 0.47, p < 0.01$ ), while coworker-reported workload was most strongly correlated with coworker-reported psychological aggression ( $r = 0.50, p < 0.01$ ). The correlations between psychological and physical aggression were generally positive. The strongest correlation was found between coworker-reported psychological and physical aggression ( $r = 0.86, p < 0.01$ ), followed by the correlation between the corresponding self-reported measures ( $r = 0.77, p < 0.01$ ).

**Table 2.** Correlations among the study variables

| No. |               | Emotional Load | Workload (S) | Workload (C) | Psy. Agg. (S) | Psy. Agg. (C) | Phy. Agg. (S) |
|-----|---------------|----------------|--------------|--------------|---------------|---------------|---------------|
| 1   | Workload (S)  | -0.34**        | 1            | -            | -             | -             | -             |
| 2   | Workload (C)  | -0.25*         | 0.56**       | 1            | -             | -             | -             |
| 3   | Psy. Agg. (S) | 0.34**         | 0.47**       | 0.28**       | 1             | -             | -             |
| 4   | Psy. Agg. (C) | 0.23**         | 0.28**       | 0.5**        | 0.28**        | 1             | -             |
| 5   | Phy. Agg. (S) | 0.30**         | 0.31**       | 0.26**       | 0.77**        | 0.28**        | 1             |
| 6   | Phy. Agg. (C) | 0.18*          | 0.24**       | 0.35**       | 0.28**        | 0.86**        | 0.3**         |

Note: \* $p < 0.05$ ; \*\* $p < 0.01$ . S = self-reported; C = coworker-reported; Psy. Agg. = psychological aggression; Phy. Agg. = physical aggression. Dashes “-” indicate values omitted from the lower-triangular correlation matrix.

#### 4.2. Moderating Role of Emotional Load in Psychological Aggression

Table 3 reports the hierarchical regression results for psychological aggression. The analyses examined whether emotional load moderated the association between workload and psychological aggression using different combinations of self-reported and coworker-reported measures.

**Table 3.** Moderating role of emotional load in the relationship between workload and psychological aggression

| Moderating Analysis | Direct Effect                 | Interaction Effect                                  | Self-Report of Psychological Aggression Behavior |         |        |              | Colleagues' Report of the Psychological Aggression |         |       |              |
|---------------------|-------------------------------|-----------------------------------------------------|--------------------------------------------------|---------|--------|--------------|----------------------------------------------------|---------|-------|--------------|
|                     |                               |                                                     | <i>b</i>                                         | $\beta$ | $R^2$  | $\Delta R^2$ | <i>b</i>                                           | $\beta$ | $R^2$ | $\Delta R^2$ |
| First               | Workload (self-report)        | -                                                   | 0.91                                             | 0.27    | 0.14   | -            | 1.71                                               | 0.36    | 0.03  | -            |
|                     | Emotional load                |                                                     | 0.33                                             | 0.26    | 0.2    | 0.05         | 0.81                                               | 0.28    | 0.07  | 0.03         |
|                     | -                             | Workload $\times$ emotional load (self-report)      | 0.16                                             | 0.56**  | 0.24** | 0.03**       | 0.18                                               | 0.22    | 0.11  | 0.03         |
| Second              | Workload (colleague - report) | -                                                   | 0.22                                             | 0.14    | 0.05   | -            | 0.24                                               | 0.08    | 0.17  | -            |
|                     | Emotional load                |                                                     | 0.37                                             | 0.24    | 0.10   | 0.04         | 0.62                                               | 0.21    | 0.26  | 0.04         |
|                     | -                             | Workload $\times$ emotional load (colleague report) | 0.12                                             | 0.33**  | 0.12** | 0.02**       | 0.3                                                | 0.63    | 0.24  | 0.02         |

Note: \*\* $p < 0.01$ . *b* = unstandardized regression coefficient;  $\beta$  = standardized regression coefficient;  $R^2$  = proportion of variance explained by the model;  $\Delta R^2$  = additional variance explained at the corresponding regression step.

“First” refers to models using self-reported workload, and “Second” refers to models using coworker-reported workload.

Dashes “-” indicate values that are not applicable.

In the model based on self-reported workload and self-reported psychological aggression, the workload  $\times$

emotional load interaction was positive and statistically significant ( $\beta = 0.56, p < 0.01$ ). Adding the interaction term increased the explained variance by 3%. This result indicates that the association between self-reported workload and self-reported psychological aggression varied according to the level of emotional load.

A significant interaction was also found when coworker-reported workload was used to predict self-reported psychological aggression ( $\beta = 0.33, p < 0.01$ ). In this model, the interaction term accounted for an additional 2% of the variance. The positive interaction coefficients in these two models were consistent with the proposed pattern in which workload had a stronger positive association with self-reported psychological aggression at higher levels of emotional load.

The interaction terms in the models predicting coworker-reported psychological aggression did not carry statistical significance indicators in Table 3. The moderating pattern was therefore supported for self-reported psychological aggression but was not consistently supported when psychological aggression was rated by coworkers. Accordingly, H1 was partially supported.

#### 4.3. Moderating Role of Emotional Load in Physical Aggression

Table 4 presents the hierarchical regression results for physical aggression. In the model using self-reported workload and self-reported physical aggression, the workload  $\times$  emotional load interaction was positive and statistically significant ( $\beta = 0.38, p < 0.01$ ). The interaction term explained an additional 1% of the variance in self-reported physical aggression.

When coworker-reported workload was used to predict self-reported physical aggression, the interaction between workload and emotional load was also positive and statistically significant ( $\beta = 0.23, p < 0.01$ ). Adding the interaction term increased the explained variance by 2%. These results indicate that the positive association between workload and self-reported physical aggression was stronger at higher levels of emotional load, regardless of whether workload was evaluated by the employee or a coworker.

The interaction terms predicting coworker-reported physical aggression did not carry statistical significance indicators in Table 4. The proposed moderating relationship was therefore supported for self-reported physical aggression but was not consistently supported for coworker-reported physical aggression. Accordingly, H2 was partially supported.

**Table 4.** Moderating role of emotional load in the relationship between workload and physical aggression

| Moderating Analysis | Direct Effect                 | Interaction Effect                                  | Self-Report of Physical Aggression Behavior |         |        |              | Colleagues' Report of the Physical Aggression |         |       |              |
|---------------------|-------------------------------|-----------------------------------------------------|---------------------------------------------|---------|--------|--------------|-----------------------------------------------|---------|-------|--------------|
|                     |                               |                                                     | <i>b</i>                                    | $\beta$ | $R^2$  | $\Delta R^2$ | <i>b</i>                                      | $\beta$ | $R^2$ | $\Delta R^2$ |
| First               | Workload (self-report)        | -                                                   | 0.65                                        | 0.24    | 0.18   | -            | 0.68                                          | 0.47    | 0.04  | -            |
|                     | Emotional load                |                                                     | 0.34                                        | 0.30    | 0.22   | 0.03         | 0.75                                          | 0.31    | 0.10  | 0.05         |
|                     | -                             | Workload $\times$ emotional load (self-report)      | 0.18                                        | 0.38**  | 0.24** | 0.01**       | 0.20                                          | 0.28    | 0.12  | 0.02         |
| Second              | Workload (colleague - report) | -                                                   | 0.25                                        | 0.14    | 0.08   | -            | 0.27                                          | 0.11    | 0.21  | -            |
|                     | Emotional load                |                                                     | 0.32                                        | 0.27    | 0.12   | 0.03         | 0.47                                          | 0.24    | 0.26  | 0.04         |
|                     | -                             | Workload $\times$ emotional load (colleague report) | 0.10                                        | 0.23**  | 0.17** | 0.02**       | 0.28                                          | 0.48    | 0.33  | 0.02         |

Note: \*\* $p < 0.01$ . *b* = unstandardized regression coefficient;  $\beta$  = standardized regression coefficient;  $R^2$  = proportion of variance explained by the model;  $\Delta R^2$  = additional variance explained at the corresponding regression step.

"First" refers to models using self-reported workload, and "Second" refers to models using coworker-reported workload.

Dashes "-" indicate values that are not applicable.

## 5. Discussion

This study examined whether emotional load moderated the relationships between workload and psychological and physical aggression among employees of an industrial firm. Workload was positively correlated with both forms of aggression, while emotional load was positively correlated with all four aggression measures. The moderation analyses showed that emotional load strengthened the associations between workload and self-reported psychological and physical aggression, whether workload was evaluated by employees or coworkers. However, the corresponding interactions were not consistently significant when aggression was evaluated by coworkers. These findings provide partial support for the proposed moderating relationships and are broadly consistent with



the stressor-emotion model, which links workplace stressors to counterproductive behavior through employees' emotional responses (Fox & Spector, 1999).

The results extend recent research that treats emotional load as a hindrance demand capable of changing how employees respond to other job demands (Machek & Machek, 2025). Rather than indicating that workload invariably leads to aggression, the findings suggest that its association with self-reported aggression depends partly on the emotional demand employees experience at the same time. One possible explanation is that employees facing high emotional load have fewer resources available to regulate their responses to workload. This interpretation is consistent with Affective Events Theory, which explains how workplace events and accumulated emotional experiences shape subsequent attitudes and behavior (Weiss & Cropanzano, 1996). It also corresponds with meta-analytic evidence identifying negative affect as an important mechanism linking workplace aggression to employee behavior and performance (Zhong et al., 2025). Negative affect and emotional regulation were not measured in the present study, however, and this explanation should therefore be regarded as a theoretically grounded interpretation rather than a tested mechanism.

The observed relationships are also consistent with previous findings on workload and workplace deviance in Middle Eastern and Asian samples (Omar et al., 2011; Radzali et al., 2013), as well as research connecting frustration with aggressive responses to workplace stressors (Bushman & Anderson, 1998). Related studies have shown that sustained emotional demands are associated with declining well-being and counterproductive behavior (Grandey, 2000; Samnani et al., 2014). A similar emotional process has been reported in research on workplace ostracism, in which anger links social exclusion to counterproductive behavior (Zhu & Zhang, 2021). Taken together, these studies suggest that emotional experiences may help explain why employees exposed to comparable workplace demands do not display the same behavioral responses. The present study adds to this literature by examining emotional load as a moderator and by distinguishing between psychological and physical aggression.

The difference between self-reported and coworker-reported outcomes warrants attention. The interaction effects were significant when aggression was reported by employees but were not consistently significant when it was evaluated by coworkers. Psychological aggression may include behavior that is not readily observable to others, while physical aggression may occur too infrequently for coworkers to provide a complete assessment. Employees and coworkers may also use different standards when evaluating workload and aggressive conduct. In addition, because emotional load and self-reported aggression were provided by the same respondents, shared measurement source may have contributed to the stronger relationships in the self-report models. The use of coworker reports remains valuable, but the inconsistent results across reporting sources indicate that the proposed moderation effect should not be treated as uniformly established.

The interaction terms accounted for an additional 1%–3% of the variance in workplace aggression, corresponding to Cohen's  $f^2$  values ranging from 0.010 to 0.031. Under Cohen (1988)'s conventions, these values represent small effects. Small interaction effects are common in organizational moderation research because behavioral outcomes are usually shaped by several individual, interpersonal, and organizational factors. Nevertheless, the limited incremental variance means that the practical importance of emotional load should not be overstated. The findings identify one condition associated with variation in employee aggression rather than offering a complete explanation of aggressive behavior at work.

From a managerial perspective, the results suggest that workload and emotional load should be considered together. Redistributing tasks may have limited relevance if employees continue to face emotionally demanding interactions, interpersonal conflict, feelings of personal threat, or persistent emotional display requirements. Related evidence shows that emotional labor and exposure to workplace violence are jointly associated with productivity loss through absenteeism and presenteeism (Ko et al., 2024). Managing emotional demands should therefore form part of a broader approach to employee well-being, work design, and workplace aggression prevention.

The findings may be particularly relevant when industrial organizations introduce new technologies or reorganize work processes. Digital systems, automated production, and electronic performance monitoring can alter task demands, work pace, and interpersonal coordination. These technological conditions were not measured in the present study and should not be interpreted as causes of the observed relationships. They nevertheless represent settings in which organizations should monitor whether changes in workload are accompanied by greater emotional pressure.

Based on the findings, five practical measures may be considered:

(1) Assess workload and emotional load together. Organizations should evaluate not only the volume and pace of assigned work but also the emotional demands accompanying that work. Separate assessments may overlook employees who face moderate task demands but high interpersonal or emotional pressure.

(2) Monitor job demands during organizational and technological change. New systems and working procedures may temporarily increase workload, uncertainty, and interpersonal tension. Managers should examine these conditions during implementation rather than assuming that new technologies immediately reduce employee demands.

(3) Train supervisors to recognize emotional overload. Supervisors should be able to identify persistent

emotional pressure, conflict, withdrawal, and escalating hostility within their teams. Early recognition allows workplace concerns to be addressed before they develop into more serious behavioral problems.

(4) Avoid simultaneous peaks in task and emotional demands. Where operational conditions permit, organizations should avoid scheduling highly demanding tasks during periods already characterized by conflict, uncertainty, or intensive interpersonal contact.

(5) Provide emotional regulation and social-support resources. Conflict-management training, access to counseling, supervisor support, and opportunities for employees to discuss difficult workplace interactions may help them manage emotional demands more effectively.

## 6. Conclusion

This study examined emotional load as a boundary condition in the relationships between workload and psychological and physical aggression among employees of an Iranian industrial firm. Workload was positively correlated with both forms of workplace aggression. The moderation analyses further showed that higher emotional load strengthened the associations between workload and self-reported psychological and physical aggression. This pattern was observed whether workload was evaluated by employees or coworkers, but it was not consistently found when aggression itself was evaluated by coworkers. Emotional load therefore appears to shape the workload–aggression relationship, although the evidence is stronger for employees’ assessments of their own behavior.

The findings contribute to organizational behavior research by showing that task and emotional demands should not be considered only as independent predictors of employee behavior. Their joint relationship may provide a more precise account of why employees facing similar workloads report different behavioral responses. By examining psychological and physical aggression separately and incorporating two reporting sources, the study also shows that conclusions concerning workplace aggression can depend on who evaluates the behavior.

The results are relevant to contemporary industrial organizations, including those undergoing digitalization and technological change. Automated systems, real-time performance monitoring, and new forms of technology-mediated coordination may alter both workload and emotional demands. Because these technological factors were not directly measured, the present findings cannot establish that digitalization or automation increases workplace aggression. Instead, they indicate that organizations introducing new technologies should monitor the task and emotional demands accompanying such changes.

Several limitations should be considered. First, the cross-sectional design does not permit causal conclusions or establish the direction of the observed relationships. Longitudinal research is needed to determine whether workload and emotional load precede changes in workplace aggression over time. Second, the participants were selected through convenience sampling from a single Iranian industrial firm, limiting the generalizability of the findings to other organizations, industries, and cultural settings. Third, emotional load was measured only through self-reports, and the stronger findings in the self-reported aggression models may partly reflect shared method variance. Coworker reports may also have been affected by differences in contact frequency and opportunities to observe aggressive behavior. Fourth, the study did not directly measure negative affect, emotional regulation, technostress, or other mechanisms that might explain the interaction between workload and emotional load. Finally, the interaction terms explained only 1%–3% of additional variance, indicating that their practical effects were small.

Future research could use longitudinal or multilevel designs, recruit participants from several organizations and national contexts, and obtain evaluations from supervisors as well as coworkers. It could also examine negative affect, emotional regulation, organizational justice, supervisor support, and technostress as possible explanatory or conditional variables. Such research would clarify when emotional load is most likely to strengthen the relationship between workload and aggressive behavior.

In conclusion, emotional load is not simply an additional workplace demand. It appears to strengthen the association between workload and employees’ reports of psychological and physical aggression, although the same pattern is less consistent in coworker evaluations. Organizations seeking to reduce workplace aggression should therefore consider employees’ emotional demands alongside the amount and pace of their work. This combined approach provides a more grounded basis for work design, employee support, and aggression prevention in industrial workplaces.

## Author Contributions

Conceptualization, N.G.; methodology, H.F.; validation, H.F.; formal analysis, N.G.; investigation H.F.; resources, N.G.; data curation, N.G.; writing—original draft preparation, H.F.; writing—review and editing, N.G. All authors have read and agreed to the published version of the manuscript.

## Data Availability

The data used to support the research findings are available from the corresponding author upon request.

## Conflicts of Interest

The authors declare no conflicts of interest.

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